

# Paper I — External Adversarial Audit — Final Report

## External Adversarial Audit — Paper I (Split-Graph Fractional Triangle-Packing Theorem)

**Audit ID:** PAPER\_I\_EXT\_AUDIT\_2026-07-10

**Date:** 2026-07-10

**Auditor:** independent external AI adversarial auditor (adversarial-math-audit protocol)

**Request spec:** PAPER\_I\_EXTERNAL\_FULL\_AUDIT\_REQUEST\_SPEC.md

**Target (only):** Theorem 1.1 of \*Affine Profile Reduction for Fractional Triangle Packings in Split Graphs\* —

> For every split graph  $G$  on  $n$  vertices,  $|E(G)| - 2 \cdot v_{\text{max}}^*(G) \leq n^2/6 + n$ ,  
> where  $v_{\text{max}}^*(G)$  is the maximum fractional triangle-packing value.

**Ground truth:** 01\_manuscript/PAPER\_I\_preprint\_v1.0.{md,pdf,tex} (Theorem 1.1, Appendix C); calculation ledger 02\_validation/CALCULATION\_LEDGER\_v1.0.md (L1–L63); Lean sources 05\_formalization/lean/.

**Prior internal status:** ledger FINAL\_PASS / FROZEN / CLEAN; manuscript claims the theorem is machine-verified in Lean, sorry-free, axioms `[propext, Classical.choice, Quot.sound]`.

### 1. Executive verdict

**Verdict: PASS (intermediate AI-assurance tier — not human peer review)**

The finite fractional split-graph theorem  $|E(G)| - 2v_{\text{max}}^*(G) \leq n^2/6 + n$  is **sound** and **correctly established** at every step the auditor could check independently, and the manuscript's scope is stated honestly (no over-claim into integral packing, clique partitions, chordal graphs, or Erdős #81). Specifically:

- **Analytic proof (Gates A–G):** the two-phase packing, the exact deficit identity, the affine/concavity reduction to pure profiles, the orbit symmetrization, the three-orbit LP, the quadratic benchmark  $R(p, q)$ , and the conversion to  $n$  all reproduce exactly. Every closed-form algebraic identity of the ledger was re-derived **symbolically** and **found exact** (Script 01, 9/9). The residual-cover LP optimum equals the closed form  $\min\{U, D, H\}$  and the full  $C(p, 2)$ -variable LP equals the 3-orbit LP (orbit symmetrization lossless), and the affine reduction with uniform bound  $R(p, q) = p/2$

holds on 400 random mixed profiles with **no counterexample** (Script 02).

- **Falsification (Gate F / kill-switch):** on 622 split graphs (structured + random + adversarial) the theorem was **never violated**; the extremal family  $K_p \vee K_{2p}$  attains  $n^2/6 + n/6$  exactly, confirming the leading constant  $1/6$  is tight and the  $+n$  term is a non-optimized slack (Script 03).
- **Formal certificate (Gate H): PASS.** An **independent rebuild** in a fresh source tree (no inherited .lake cache; community Mathlib cache) reports `Build completed successfully (8027 jobs), exit 0, with PaperI.paperI\_main` **and** the LP-duality node `PaperI.Split.residual_duality` each depending on axioms `[propext, Classical.choice, Quot.sound]` — no `sorryAx`, no `native_decide`, no project-specific axiom. The released Lean sources match the recorded integrity manifest by SHA-256 (8/8), the Lean statement `paperI_main :  $\Phi \leq n^2/6 + n$  ( $\Phi = \text{edgeCount} - 2 \cdot \text{nu3star}$ ) is exactly Theorem 1.1, and a static scan finds no sorry/admit/native_decide/added axiom in the sources. This independently reproduces the recorded gate log.`
- **Citations (Gate I):** the single proof-critical import — finite-dimensional LP strong duality — is standard, and the Lean development derives it from Mathlib's finite Farkas (`residual_duality`), so the result does not depend on the exact Schrijver reference. No counterexample, circularity, unmet hypothesis, or scope over-claim was found. This is an independent AI adversarial audit: an intermediate assurance tier, stronger than self-audit but **not** external human peer review; it confers no publication status.

## 2. Scope, method, independence

**In scope:** Theorem 1.1 (finite fractional bound for split graphs) and its full proof chain, plus the Lean certificate for `paperI_main`.

**Out of scope (confirmed absent from the claim):** integral triangle packing, clique partitions/cp, chordal graphs,  $\Phi_T$ , asymptotic Erdős #81, any fractional-to-integral transfer. Paper II and the asymptotic #81 route were not consulted.

**Method:** clean-context decomposition into claim IDs mapped to gates A–J; predeclared kill-switches for every computation; independent re-implementation of all primitives ( $\forall$  LP, residual-cover LP, orbit LP, symbolic algebra); literal statement/definition cross-check across manuscript  $\leftrightarrow$  ledger  $\leftrightarrow$  Lean; independent Lean build in a fresh tree; SHA-256 provenance. **Independence:** fresh auditor code; fresh Lean source copy with no inherited .lake cache; the author's repo of record was not used. See `ENVIRONMENT.md`.

### 3. Gate table (A–J)

| Gate | Item                                                                                                                           | Status                      | Evidence                                                                                                                                                                    |
|------|--------------------------------------------------------------------------------------------------------------------------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A    | Definitions ( $v^*$ , deficit, split model, $\kappa$ , profiles, orbits) well-posed & consistent across manuscript/ledger/Lean | VERIFIED_WITHIN_SCOPE       | manuscript §2 $\leftrightarrow$ ledger L1–L4 $\leftrightarrow$ Lean Split,Phi,nu3star,edgeCount,n cross-checked                                                             |
| B    | Aggregate-load / affine profile reduction; candidates suffice                                                                  | VERIFIED_WITHIN_SCOPE       | Lemma 3.1 (L4) exact; concavity L33/L34; Script 02(B) 400 mixed profiles, 0 counterexample                                                                                  |
| C    | Orbit symmetrization + finite LP; solved independently                                                                         | VERIFIED_WITHIN_SCOPE       | Script 02: full C(p,2)-LP = 3-orbit LP = $\min\{U, D, H\}$ (0/660 & 0/280, err 5e-15)                                                                                       |
| D    | Finite LP duality applied within hypotheses; no infinite-dim appeal                                                            | VERIFIED_WITHIN_SCOPE       | App B: finite feasible bounded pair; weak duality + attainment; Lean derives from Farkas                                                                                    |
| E    | Deficit identity (4.9/L28) + quantitative bound $R(p, q)$ ; constant recomputed                                                | VERIFIED_WITHIN_SCOPE       | Script 01: L28, L48, L50, L51, L53, L58, q=0 all exact (9/9)                                                                                                                |
| F    | Conversion to $n$ ( $\leq n^2/6+n$ ) + comparison with optimum packing; tightness                                              | VERIFIED_WITHIN_SCOPE       | L58–L63 exact; Script 03: 0 violations/622, extremal attains $n^2/6+n/6$                                                                                                    |
| G    | Assembly composes to the theorem; no circularity                                                                               | VERIFIED_WITHIN_SCOPE       | linear chain L1→L63; Lean paperI_main assembles nu3star_ge_Vcom+Mcov_lower+edgeCount_eq+arithmetic                                                                          |
| H    | Lean: build, sorry-freeness, #print axioms, statement match, SHA                                                               | VERIFIED_WITHIN_SCOPE       | Script 04: independent rebuild 8027 jobs exit 0; paperI_main+residual_duality axioms [proptext,Classical.choice,Quot.sound], no sorryAx; SHA 8/8 match; statement = Thm 1.1 |
| I    | Citations (LP duality; split-graph structure) applied within hypotheses                                                        | VERIFIED_WITH_RESIDUAL_RISK | finite LP duality standard + Lean-derived; exact "Cor. 7.1g" label not book-checked                                                                                         |
| J    | Over-claim / scope check (no chordal, cp, #81, integral)                                                                       | VERIFIED_WITHIN_SCOPE       | manuscript §9, ledger §15 explicitly disclaim; statement is fractional split-graph only                                                                                     |

### 4. Independent computation (Scripts 01–03)

All scripts declared kill-switches before running; seed 20260710; artifacts (script, results.txt, report.pdf, .zip, .sha256) under scripts/.

- **01 — algebraic identities (Gate E).** sympy exact check of L48, L50, L51 (SOS), L53, L58, the q=0 facts, and L25/L28. **9/9 exact** → CALCULATION\_VERIFIED.
- **02 — orbit LP / symmetrization / reduction (Gates B, C).** (C0) q=0 branch verified; (C1) orbit LP =  $\min\{U, D, H\}$  (0/660); (C2) full LP = orbit LP, err 5e-15 (0/280) — orbit averaging is lossless; (B)  $M(\kappa) \geq \min_s M(\kappa^s)$  and  $M(\kappa) \geq R-p/2$  on 400 mixed profiles, 0 violations, bound tight (slack 0). Auditor correction CORR-02-01 recorded

(a  $q=0$  domain over-reach in the first run; not a paper defect).

- **03 — split-graph falsification (Gate F).** 622 split graphs, 0 theorem

violations; max ratio  $(|E| - 2v_{\min}^*) / (n^2/6 + n) = 0.815$ ; extremal  $K_{p \vee K_{\lfloor 2p \rfloor}}$  attains  $n^2/6 + n/6$  exactly (0 tightness failures).

## 5. Falsification / kill-switch log

No counterexample was found to Theorem 1.1 or to any gate. Failed falsification attempts (supporting evidence, not proof): adversarial dense complete-split graphs and 600 random split graphs (Script 03); random mixed neighborhood profiles probing the affine reduction and the uniform bound (Script 02); a search for an orbit/candidate beating  $\min\{U, D, H\}$  (Script 02, none). The one issue surfaced was an **auditor** domain error (CORR-02-01), not a defect in the paper. Bounded search is existential evidence; the universal statement rests on the analytic proof + the Lean certificate.

## 6. Coverage declaration

**Verified (within scope):** all algebraic identities of the ledger (exact); the orbit LP closed form and the losslessness of orbit symmetrization (LP, bounded  $p$ ); the affine reduction and uniform bound (LP, bounded  $p$ , random mixed profiles); the end-to-end inequality and extremal tightness (bounded split graphs); definition consistency across manuscript/ledger/Lean; Lean statement = Theorem 1.1; source SHA-256 = recorded manifest; static sorry/axiom-cleanliness of the sources; scope non-over-claim.

**Residual risks:** (i) numeric LP/enumeration checks are over bounded ranges ( $p \leq 11$  for the LP identities,  $p \leq 7-8$  for exact  $v_{\min}^*$ ), so they corroborate rather than prove the universal statement — but the analytic proof (Gates A–G) and the Lean certificate (Gate H) carry the universal claim; (ii) the exact wording of Schrijver "Corollary 7.1g" was not verified against the physical book (immaterial: the duality is standard and Lean-derived from Farkas); (iii) this is an AI audit, not human peer review.

## 7. Recommendation

Maintain the theorem at its internal FINAL\_PASS / FROZEN status for the **\*\*finite fractional split-graph bound only\*\***. The independent audit found the analytic proof, the computational checks, and the formal certificate mutually consistent and free of any detected error, circularity, or over-claim. Human expert / journal peer review remains the appropriate next tier; this AI audit confers no publication status.

\*Distinctions maintained: "analytic proof correct" (Gates A–G, I), "Lean certificate holds" (Gate H), and "computation found no counterexample" (Scripts 02–03) are reported separately; all three point the same way for Paper I.\*